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### Camera time-based motion trails and motion heat-maps for periodic captured images

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#### Abstract

Disclosed herein are system, method and/or computer program product embodiments, and/or combinations and sub-combinations thereof, for generating a long exposure composite image (LECI) frame, wherein the LECI frame provides a summary of activity in a monitored environment over a duration of time. An embodiment captures, using an image-capturing device, a sequence of image frames over the duration of time using an optical sensor of the image-capturing device. The embodiment then generates the LECI frame, using an embedded low-power processor coupled to the optical sensor and dedicated to generating LECI frames, by combining the sequence of image frames. The embodiment then transmits the LECI frame to a network device using a transceiver coupled to the low-power processor.

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**Background/Summary**

BACKGROUND

Field  
(1) This disclosure is generally directed to image processing, and more particularly to generating a composite image that represents a visual summary of captured images and that may include camera time-based motion trails and heat-maps based on the captured images.

SUMMARY

(2) Provided herein are system, apparatus, article of manufacture, method and/or computer program

product embodiments, and/or combinations and sub-combinations thereof, for generating long exposure composite image (LECI) frames, where the LECI frames provide a single-image visual summary of activity in a monitored environment over a duration of time.

(3) Some aspects of this disclosure relate to a method for generating a LECI frame. According to some aspects, the method includes capturing a sequence of image frames of view of a monitored environment over the duration of time using an optical sensor of the image-capturing device. In some aspects, this step may be performed by enabling the optical sensor to be always recording the monitored environment. The method further includes generating the LECI frame using an embedded low-power processor coupled to the optical sensor and dedicated to generating LECI frames by combining the sequence of image frames. The embedded low-power processor may be configured to operate separately from a main processor of the image-capturing device and may even be configured to operate if the main processor is in a sleep or low-power mode. The generated LECI frame may then be transmitted to a network device using a transceiver coupled to the embedded low-power processor.

(4) According to some aspects, image frames within the sequence of image frames may be weighted based on the time period in which the image frames were captured. For example, the sequence of image frames includes a first set of image frames and a second set of image frames, where the first set of image frames is associated with a first time period within the duration of time and the second set of image frames are associated with a different time period within the duration of time. The first time period is more recent than the second time period, and generating the composite image frame may include performing weighted averaging on the image frames within the sequence. Weighted averaging may include, for example, assigning a first set of weights to the first set of image frames of the sequence of image frames and a second set of weights to the second set of image frames, where the first set of weights is greater than the second set of weights.

(5) According to some aspects, the method further includes generating the LECI frame by combining the sequence of image frames by computing a pixel-wise weighted average of pixel values across the sequence of image frames based on the first and second set of weights. According to some aspects, the method further includes generating the LECI by performing image detection on one or more image frames. Image detection may be used, for example, to identify a moving object in one or more image frames of the sequence of image frames, and performing the weighted averaging by assigning higher weights to the pixels corresponding to the moving object in the one or more image frames. According to some aspects, the method further includes generating the LECI frame by identifying the relative displacement of one or more pixels between a set of image frames of the sequence of image frames to generate one or more annotations in the LECI frame. Some examples of annotations include direction vectors, such as arrows, indicating the direction of motion of an object in the sequence of image frames and heat maps indicating the length of time that an object stayed in a particular location in the monitored environment.

(6) According to some aspects, the method further includes capturing the sequences of image frames at a capture rate that is proportional to a measure of activity in the monitored environment. According to some aspects, the duration of time is inversely proportional to a measure of activity in the monitored environment. According to some aspects, the duration of time is inversely proportional to a measure of activity in the monitored environment.

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## Description

### BRIEF DESCRIPTION OF THE FIGURES

(1) The accompanying drawings are incorporated herein and form a part of the specification.

(2) FIG. 1 illustrates a block diagram of a multimedia environment, according to some embodiments.

- (3) FIG. 2 illustrates a block diagram of a streaming media device, according to some embodiments.
- (4) FIG. 3 illustrates a block diagram of an example image-capturing device for generating long exposure composite image (LECI) frames, according to some embodiments.
- (5) FIG. 4A illustrates an exemplary LECI frame, according to some embodiments.
- (6) FIG. 4B illustrates another exemplary LECI frame, according to some embodiments.
- (7) FIG. 5 illustrates a flow diagram of a method for generating LECI frames from stored video data, according to some embodiments.
- (8) FIG. 6 illustrates a flow diagram of a method for generating LECI frames from a sequence of image frames, according to some embodiments.
- (9) FIG. 7 illustrates a flow diagram of a method for identifying the movement of large objects between frames of video data, according to some embodiments.
- (10) FIG. 8 illustrates an example computer system useful for implementing various embodiments.
- (11) In the drawings, like reference numbers generally indicate identical or similar elements. Additionally, generally, the left-most digit(s) of a reference number identifies the drawing in which the reference number first appears.

#### DETAILED DESCRIPTION

- (12) Provided herein are system, apparatus, article of manufacture, method and/or computer program product embodiments, and/or combinations and sub-combinations thereof, for generating long exposure composite image (LECI) frames, where a LECI provides a single-image visual summary of activity that occurred over a duration of time. For example, aspects herein describe capturing a sequence of image frames at a low frame rate and generating a LECI frame by combining the sequence of image frames.
- (13) Surveillance cameras capture and process images and video on limited battery power over extended periods. Maximizing the battery life of such surveillance cameras is crucial for their effectiveness and longevity. Cameras operating at a low frame rate generally consume less power and generate less data, leading to reduced storage costs. However, lower frame rates may result in choppy video, especially for fast-moving subjects. Hence, the frame rate of a surveillance camera involves a tradeoff between capturing smooth video and conserving power and storage resources.
- (14) Furthermore, reviewing recorded video from a surveillance camera over extended periods can be challenging. Using tools such as increased playback speed, timestamped logs, and motion detection alerts may not be the most efficient video reviewing approaches, especially when the goal is to quickly identify relevant events while minimizing time spent on irrelevant footage.
- (15) Aspects of this disclosure describe an efficient way to summarize activity that has occurred over a duration of time in a monitored environment. According to some aspects, LECI frames are generated using a sequence of image frames captured over a duration of time. Each LECI frame summarizes activity that has occurred over a duration of time in a monitored environment using a single-image, resulting in reduced bandwidth and storage demands. According to some aspects, a LECI frame is generated by combining a sequence of image frames by computing a pixel-wise weighted average of pixel values across the sequence of image frames.
- (16) Various aspects of this disclosure may be implemented using and/or may be part of a multimedia environment **102** shown in FIG. 1. It is noted, however, that multimedia environment **102** is provided solely for illustrative purposes, and is not limiting. Embodiments of this disclosure may be implemented using and/or may be part of environments different from and/or in addition to the multimedia environment **102**, as will be appreciated by persons skilled in the relevant art(s) based on the teachings contained herein. An example of the multimedia environment **102** shall now be described.
- (17) Multimedia Environment
- (18) FIG. 1 illustrates a block diagram of a multimedia environment **102**, according to some embodiments. In a non-limiting example, multimedia environment **102** may be directed to

streaming media. However, this disclosure is applicable to any type of media (instead of or in addition to streaming media), as well as any mechanism, means, protocol, method and/or process for distributing media.

(19) Multimedia environment **102** may include one or more media systems **104**. A media system **104** could represent a family room, a kitchen, a backyard, a home theater, a school classroom, a library, a car, a boat, a bus, a plane, a movie theater, a stadium, an auditorium, a park, a bar, a restaurant, or any other location or space where it is desired to receive and play streaming content. User(s) **132** may operate with the media system **104** to select and consume content.

(20) Each media system **104** may include one or more media devices **106** each coupled to one or more display devices **108**. It is noted that terms such as “coupled,” “connected to,” “attached,” “linked,” “combined” and similar terms may refer to physical, electrical, magnetic, logical, etc., connections, unless otherwise specified herein.

(21) Media device **106** may be a streaming media device, DVD or BLU-RAY device, audio/video playback device, cable box, and/or digital video recording device, to name just a few examples. Display device **108** may be a monitor, television (TV), computer, smart phone, tablet, wearable (such as a watch or glasses), appliance, internet of things (IoT) device, and/or projector, to name just a few examples. In some embodiments, media device **106** can be a part of, integrated with, operatively coupled to, and/or connected to its respective display device **108**. In some embodiments, image capturing device **134** may be operatively coupled to, and/or connected to media system **104** and communicate to content server(s) **120** and/or system server(s) **126** via media system **104**. In some aspects, image-capturing device **134** may communicate directly with content server(s) **120** and/or system server(s) **126** without needing to communicate via media system **104**.

(22) Each media device **106** may be configured to communicate with network **118** via a communication device **114**. Communication device **114** may include, for example, a cable modem or satellite TV transceiver. Media device **106** may communicate with communication device **114** over a link **116**, wherein link **116** may include wireless (such as Wi-Fi) and/or wired connections.

(23) In various embodiments, network **118** can include, without limitation, wired and/or wireless intranet, extranet, Internet, cellular, Bluetooth, infrared, and/or any other short range, long range, local, regional, global communications mechanism, means, approach, protocol and/or network, as well as any combination(s) thereof.

(24) Media system **104** may include a remote control **110**. Remote control **110** can be any component, part, apparatus and/or method for controlling media device **106** and/or display device **108**, such as a remote control, a tablet, laptop computer, smartphone, wearable, on-screen controls, integrated control buttons, audio controls, or any combination thereof, to name just a few examples. In an embodiment, remote control **110** wirelessly communicates with media device **106** and/or display device **108** using cellular, Bluetooth, infrared, etc., or any combination thereof. Remote control **110** may include a microphone **112**, which is further described below.

(25) Multimedia environment **102** may include a plurality of content servers **120** (also called content providers, channels or sources). Although only one content server **120** is shown in FIG. 1, in practice multimedia environment **102** may include any number of content servers **120**. Each content server **120** may be configured to communicate with network **118**.

(26) Each content server **120** may store content **122** and metadata **124**. Content **122** may include any combination of music, videos, movies, TV programs, multimedia, images, still pictures, text, graphics, gaming applications, advertisements, programming content, public service content, government content, local community content, software, and/or any other content or data objects in electronic form.

(27) In some embodiments, metadata **124** comprises data about content **122**. For example, metadata **124** may include associated or ancillary information indicating or related to writer, director, producer, composer, artist, actor, summary, chapters, production, history, year, trailers, alternate versions, related content, applications, and/or any other information pertaining or relating to

content **122**. Metadata **124** may also or alternatively include links to any such information pertaining or relating to content **122**. Metadata **124** may also or alternatively include one or more indexes of content **122**.

(28) Multimedia environment **102** may include one or more system servers **126**. System servers **126** may operate to support media devices **106** from the cloud. It is noted that the structural and functional aspects of system servers **126** may wholly or partially exist in the same or different ones of system servers **126**.

(29) The media devices **106** may exist in thousands or millions of media systems **104**. Accordingly, the media devices **106** may lend themselves to crowdsourcing embodiments and, thus, the system servers **126** may include one or more crowdsource servers **128**.

(30) For example, using information received from the media devices **106** in the thousands and millions of media systems **104**, the crowdsource server(s) **128** may identify similarities and overlaps between closed captioning requests issued by different users **132** watching a particular movie. Based on such information, the crowdsource server(s) **128** may determine that turning closed captioning on may enhance users' viewing experience at particular portions of the movie (for example, when the soundtrack of the movie is difficult to hear), and turning closed captioning off may enhance users' viewing experience at other portions of the movie (for example, when displaying closed captioning obstructs critical visual aspects of the movie). Accordingly, the crowdsource server(s) **128** may operate to cause closed captioning to be automatically turned on and/or off during future streaming of the movie.

(31) The system servers **126** may also include an audio command processing module **130**. As noted above, remote control **110** may include microphone **112**. Microphone **112** may receive audio data from users **132** (as well as other sources, such as the display device **108**). In some embodiments, media device **106** may be audio responsive, and the audio data may represent verbal commands from user **132** to control media device **106** as well as other components in media system **104**, such as display device **108**.

(32) In some embodiments, the audio data received by microphone **112** in remote control **110** is transferred to media device **106**, which then forwards the audio data to audio command processing module **130** in system servers **126**. Audio command processing module **130** may operate to process and analyze the received audio data to recognize a verbal command of user **132**. Audio command processing module **130** may then forward the verbal command back to media device **106** for processing.

(33) In some embodiments, the audio data may be alternatively or additionally processed and analyzed by an audio command processing module **216** in media device **106** (see FIG. 2). Media device **106** and system servers **126** may then cooperate to pick one of the verbal commands to process (either the verbal command recognized by audio command processing module **130** in system servers **126**, or the verbal command recognized by audio command processing module **216** in media device **106**).

(34) FIG. 2 illustrates a block diagram of an example media device **106**, according to some embodiments. Media device **106** may include a streaming module **202**, a processing module **204**, storage/buffers **208**, and a user interface module **206**. As described above, user interface module **206** may include audio command processing module **216**.

(35) Media device **106** may also include one or more audio decoders **212** and one or more video decoders **214**.

(36) Each audio decoder **212** may be configured to decode audio of one or more audio formats, such as but not limited to AAC, HE-AAC, AC3 (Dolby Digital), EAC3 (Dolby Digital Plus), WMA, WAV, PCM, MP3, OGG GSM, FLAC, AU, AIFF, and/or VOX, to name just some examples.

(37) Similarly, each video decoder **214** may be configured to decode video of one or more video formats, such as but not limited to MP4 (mp4, m4a, m4v, f4v, f4a, m4b, m4r, f4b, mov), 3GP (3gp,

3gp2, 3g2, 3gpp, 3gpp2), OGG (ogg, oga, ogv, ogx), WMV (wmv, wma, asf), WEBM, FLV, AVI, QuickTime, HDV, MXF (OP1a, OP-Atom), MPEG-TS, MPEG-2 PS, MPEG-2 TS, WAV, Broadcast WAV, LXF, GXF, and/or VOB, to name just some examples. Each video decoder **214** may include one or more video codecs, such as but not limited to H.263, H.264, H.265, AVI, HEV, MPEG1, MPEG2, MPEG-TS, MPEG-4, Theora, 3GP, DV, DVCPRO, DVCProHD, IMX, XDCAM HD, XDCAM HD422, and/or XDCAM EX, to name just some examples.

(38) Streaming module **202** of media device **106** may be configured to receive image information from image capturing device **134**. In some aspects, the image information may comprise LECI frame generated by a low-power processor of the image-capturing device **134**. In some aspects, the image information may comprise a sequence of image frames recorded by the image-capturing device **134** and an indication (e.g., a flag, a bit in a header of a packet) that the media device **106** can generate a LECI frame from the provided image information. For example, processing module **204** may receive the sequence of image frames from image capturing device **134** and generate a LECI frame from the provided sequence. In this manner, image-capturing device **134** may offload LECI processing to the media device **106**. For example, image-capturing device **134** may determine it lacks sufficient processing power or electrical power (e.g., a low battery) to generate LECI frames and, instead, transmits the recorded sequence of image frames to media device **106**.

(39) Now referring to both FIGS. 1 and 2, in some embodiments, user **132** may interact with media device **106** via, for example, remote control **110**. For example, user **132** may use remote control **110** to interact with user interface module **206** of media device **106** to select a content item, such as a movie, TV show, music, book, application, game, etc. In response to the user selection, streaming module **202** of media device **106** may request the selected content item from content server(s) **120** over network **118**. Content server(s) **120** may transmit the requested content item to streaming module **202**. Media device **106** may transmit the received content item to display device **108** for playback to user **132**.

(40) In some aspects, media device **106** may display an interface for interacting with the sequence of image frames provided by image capturing device **134**. For example, the interface may display selectable options for generating LECI frames based on the sequence of image frames. One example of a selectable option is the duration of time (e.g., 1 minute, 5 minutes) of the sequence of images for which to generate the LECI images. Another example includes the types of annotations or effects (e.g., arrows, heat maps, highlighting, blurring) to be added to the LECI to represent actions or objects detected within the frames of the sequence of frames.

(41) In streaming embodiments, streaming module **202** may transmit the content item to display device **108** in real time or near real time as it receives such content item from content server(s) **120**. In non-streaming embodiments, media device **106** may store the content item received from content server(s) **120** in storage/buffers **208** for later playback on display device **108**.

(42) Generating Long Exposure Composite Image Frames

(43) FIG. 3 illustrates a block diagram of an example image-capturing device **300** for generating long exposure composite image (LECI) frames, according to some aspects of this disclosure. According to some aspects, image-capturing device **300** (which can be image-capturing device **134** of FIG. 1) may be configured to communicate with media system **104**, content server(s) **120**, or system server(s) **126** in multimedia environment **102** of FIG. 1. According to some aspects, image-capturing device **300** may be a security camera, closed-circuit television camera, digital camera, smartphone, webcam, camcorder, medical imaging devices, and thermal cameras, to name just some examples.

(44) In the illustrated example, image-capturing device **300** includes an optical sensor **302**, an image processor **306**, a video encoder **312**, central processor **314**, a storage module **308**, battery module **310**, and a transceiver. Optical sensor **302** may include an embedded low-power processor **304** dedicated to generating LECIs.

(45) According to some aspects, optical sensor **302** may be configured to capture images and/or

video of a monitored environment. Optical sensor **302** may have two components: a light-sensitive pixel array that converts photons to electric charges and a read-out logic that converts charges to digital values (i.e., raw image pixels). According to some aspects, optical sensor **302** may be a charge-coupled device (CCD) sensor that uses an arrangement of capacitors to move the electrical charges from pixel to pixel until they are read out from the sensor. Alternatively, optical sensor **302** may be a complementary metal-oxide semiconductor (CMOS) sensor that uses transistors at each pixel location to amplify and read out the electrical charges.

(46) According to some aspects, optical sensor **302** generates uncompressed raw pixel data corresponding to a captured image and/or video. According to some aspects, optical sensor **302** may capture color information by using a pattern of red, green, and blue (RGB) filters over the pixels. According to some aspects, optical sensor **302** may capture luminance (Y) and chrominance (UV) components of the pixels. The uncompressed raw pixel data generated by the optical sensor **302** may then be processed by the embedded low-power processor **304** and/or image processor **306** of the image-capturing device **300**.

(47) According to some aspects, embedded low-power processor **304** is a specialized integrated circuit designed to provide processing capability for specific tasks (e.g., computing the weighted average of a set of image frames) so as to conserve power consumption by image-capturing device **300**. Embedded low-power processor **304** may minimize power consumption through a combination of architectural design and power management techniques. For example, embedded low-power processor **304** may use energy-efficient microarchitectures to reduce power consumption. Furthermore, embedded low-power processor **304** may operate at a lower clock speed and voltage compared to the main processor (i.e., image processor **306**) of the image-capturing device **300**, and the slower clock speeds and reduced voltage levels result in lower power consumption.

(48) According to some aspects, raw image and/or video pixels are transferred from the optical sensor **302** (e.g., via a camera serial interface) to image processor **306**. Image processor **306** may perform image compression by generating quantized pixels. Furthermore, image processor **306** may also perform noise reduction, color correction, gamma correction, and other enhancements to improve the overall quality of the image.

(49) According to some aspects, operations performed by the image processor **306** may consume substantial amount of battery power. Therefore, embedded low-power processor **304** may be configured to operate when image-capturing device **300** is in a power state for conserving power consumption (e.g., low power or sleep state). Operations (e.g., generating a LECI frame) of embedded low-power processor **304** are configured to be substantially power-efficient compared to operations using the main processor (i.e., image processor **306**). To reduce power consumption, image processor **306** may implement sleep cycles. A sleep cycle may be designed to reduce power consumption during periods of inactivity while still allowing the sensor to quickly resume normal operation when needed.

(50) According to some aspects, the video encoder **312** takes raw video data from optical sensor **302** and converts it into a digital format that can be stored or transmitted over a network. Video encoder **312** may be embedded on a system on chip (SoC) of image capturing device **300**.

According to some aspects, video encoder **312** may use video compression algorithms to reduce the size of the video data. Common video compression standards include H.264, H.265 (high efficiency video coding (HEVC)), and VP9. According to some aspects, to reduce power consumption, instead of central processor **314**, video encoder **312** may be configured to generate motion vectors from a sequence of image frames of the input video data. In other words, video encoder **312** may be configured to perform frame-by-frame comparison when compressing the video. Video encoder **312** provides image capturing device **300** low-power hardware acceleration because the generation of motion vectors is off-loaded to the video encoder **312**. Motion vectors may be used to describe how objects in a video move from one frame to another. The output of



video encoder **312** may be a compressed video bitstream. Central processor **314** may scan the generated bitstream for the motion vectors created by video encoder **312**. In some aspects, video encoder **312** may be built into a system-on-a-chip of image capturing device **300**.

(51) According to some aspects, LECI frames generated by embedded low-power processor **304** may be stored in storage module **308**. Additionally, image and video data generated by image processor **306** may also be stored in storage module **308**. Storage module **308** may be a built-in internal memory that can store a limited number of images and videos. According to some aspects, the storage module **308** may include a removable memory card such as Secure Digital (SD), microSD, CompactFlash (CF), and the like.

(52) According to some aspects, LECI frames generated by the embedded low-power processor **304** are stored in the storage module **308** and may be periodically transmitted to media device **104** and/or uploaded to a cloud storage system of multimedia environment **102**. User **132** can download LECIs to review activity in the monitored environment over the corresponding durations of time corresponding to the LECI frames. According to some aspects, image-capturing device may transmit image and video data to media device **104** and/or the cloud storage system for generation of the LECI frames.

(53) According to some aspects, when the image-capturing device **300** is power-constrained (e.g., low power state, sleep state, low battery), image-capturing device **300** may not upload LECI frames to the cloud. Instead, image capturing device **300** may provide an interface for direct access of the LECIs from the storage module **308** over a local network (e.g., user WiFi network). According to some aspects, when the image-capturing device **300** is not power-constrained, image-capturing device **300** may capture a video of the monitored environment over a duration of time, and upload the video data to media device **104** and/or a cloud storage system of multimedia environment **102**. According to some aspects, a LECI generation module **136** may generate one or more LECIs corresponding to the video data. The cloud storage system of multimedia environment **102** may transmit the LECI frames to media system(s) **102** which may provide an interface for reviewing the LECIs and determining the corresponding activity in the monitored environment over the corresponding durations of time.

(54) As noted above, optical sensor **302** captures a sequence of image frames at a predetermined frame rate over a duration of time, and the embedded low-power processor **304** or the image processor **306** combines the sequence of images to generate an LECI frame. According to some aspects, the frame rate at which the image frames are captured by the optical sensor **302** may be predetermined based on a measure of activity in the monitored environment. When monitoring a dynamic environment with a large amount of activity, optical sensor **302** of the image-capturing device **300** may be configured to record image frames at a higher frame rate to ensure that the sequence of captured image frames captures a sufficient amount of motion. Alternatively, optical sensor **302** of the image-capturing device **300** may be configured to record image frames at a lower frame rate when monitoring an environment with a low amount of activity to ensure that the sequence of captured image frames captures a sufficient amount of motion. Image capturing device **300** may be configured with software to perform motion detection across the sequence of captured image frames.

(55) According to some aspects, image-capturing device **300** may quantify a measure of activity in the monitored environment based on comparing two or more of the captured images. For example, image processor **306** of image capturing device **300** may be configured to quantify activity in the monitored environment using techniques such as optical flow method, motion histogram, average displacement vector, mean squared error technique, and the like.

(56) According to some aspects, when image-capturing device **300** determines that a measure of activity corresponding to the monitored environment exceeds a predetermined activity threshold, optical sensor **302** may be configured to capture the image frames at a first predetermined rate. Similarly, when image-capturing device **300** determines that the measure of activity is below the

predetermined activity threshold, optical sensor **302** may be configured to capture the image frames at second predetermined rate that is lower than the first predetermined rate. That is the first determined rate (e.g., more than 30 image frames per minute) may be higher than the second predetermined rate (e.g., less than or equal to 30 image frames per minute).

(57) As noted above, each LECI frame provides a single-image visual summary of activity in a monitored environment over a duration of time. In some aspects, image capturing device **300** is recording images of the monitored environment in a fixed position such that the single-image visual summary of activity represents a summary of the activity of the monitored environment from a single view. According to some aspects, the duration of time corresponding to each LECI frame (i.e., the period of activity summarized by each LECI frame) may be varied based on a measure of activity in the monitored environment and/or based on selectable user options. In some aspects, the duration of time corresponding to each LECI frame may determine the rate at which the LECI frames are generated by the image-capturing device **300**. For example, the rate at which LECI frames are generated may be varied based on a measure of activity in the monitored environment.

(58) According to some aspects, image-capturing device **300** may determine it is monitoring a dynamic environment with a large amount of activity. Image capturing device **300** may be configured to generate LECI frames that correspond to a shorter duration of time. Such a function may ensure that each LECI frame provides an uncluttered summary of activity over the respective duration of time. Furthermore, when monitoring an environment with a low amount of activity, generating LECI frames that correspond to a longer duration of time may be sufficient to ensure an uncluttered summary of activity by each LECI frame over the respective duration of time.

(59) According to some aspects, when image-capturing device **300** determines that a measure of activity corresponding to the monitored environment exceeds a predetermined activity threshold, embedded low-power processor **304** may be configured to generate LECI frames that correspond to a shorter duration of time (e.g., each LECI frame summarizing less than 4 minutes of activity) than when the measure of activity is less than the predetermined activity threshold. Similarly, when image-capturing device **300** determines that a measure of activity corresponding to the monitored environment falls below a predetermined activity-threshold, embedded low-power processor **304** may be configured to generate LECI frames that correspond to a longer duration of time (e.g., each LECI frame summarizing more than 4 minutes of activity).

(60) FIG. 4A illustrates an exemplary long exposure composite image (LECI), according to some aspects of this disclosure. The example LECI frame **400A** of FIG. 4A provides a single-image visual summary of a dog running across a predominantly static background (e.g., mountains **406** and trees **408**) over a period of time.

(61) As noted above, the embedded low-power processor **304** and/or the image processor **306** may generate the example LECI frame by combining a sequence of image frames that are captured by the optical sensor **302**. According to some aspects, to generate a LECI frame, the embedded low-power processor **304** first aligns the sequence of image frames captured by optical sensor **302** before combining them to generate the LECI frame. According to some aspects, the sequence of image frames may be combined using pixel-wise weighted averaging and/or stacking methods. Furthermore, the sequence of image frames comprises a set of image frames captured in succession over a period of time. The weighted averaging may be performed over a subset, or over all, of the image frames of the sequence of image frames.

(62) According to some aspects, computing the pixel-wise weighted average of a set of image frames may involve computing a weighted average pixel value for each pixel location over all image frames of the set of image frames. According to some aspects, optical sensor **302** may capture image frames in an RGB format comprising red, green, and blue channels. Computing the weighted average of a set of RGB image frames may involve computing a pixel-wise weighted average for each color channel.

(63) According to some aspects, optical sensor **302** may capture image frames in a format that

indicates the luminance and chrominance values of each pixel (e.g., YUV, YCbCr, etc.). Computing the weighted average of such image frames may involve computing a pixel-wise weighted average of luminance values across a set of image frames while ignoring the chrominance value.

Alternatively or in addition, computing the weighted average of image frames may involve computing a pixel-wise weighted average of chrominance values across the set of image frames. (64) According to some aspects, when computing the weighted average of a sequence of image frames, embedded low-power processor **304** may select the weights assigned to the pixels to accentuate one or more moving objects in the monitored environment. Alternately, or in addition, image processor **306** may generate a LECI frame by computing the weighted average of the sequence of image frames. According to some aspects, the embedded low-power processor **304** and/or the image processor **306** may determine that they lack sufficient processing power and/or electrical power (i.e., due to a low battery) to generate a LECI frame and, instead may transmit the sequence of image frames to media device **106**. In such a case, the media device **106** may receive the sequence of image frames and generate a LECI frame by computing the weighted average of the sequence of image frames. In some aspects, steps for generating a LECI frame may be performed by embedded low-power processor **304**; in some aspects, the steps may be distributed between different devices such as one or more of the embedded low-power processor, media device **106**, content server(s) **120**, and system server(s) **126**.

(65) According to some aspects, the LECI frame may be generated with annotations to represent actions performed by the one or more moving objects. For example, when performing pixel-wise weighted averaging across a sequence of image frames, newer pixels may be assigned higher weights compared to older pixels to obtain a streak effect in the direction of motion in the LECI frame. According to some aspects, when combining a sequence of image frames to generate a LECI frame, pixel colors may be modified according to the pixel age such that newer image frames of the sequence may have a color and/or a shade of color that is different from older image frames. According to some aspects, when combining a sequence of image frames to generate a LECI frame, pixel colors may be modified to create a heat map of activity in the LECI frame. According to some aspects, creating a LECI frame that displays a heat map of activity may involve maintaining pixel-change histograms to indicate regions where the most motion has occurred and making them a different color (e.g., using shades of red for high activity areas).

(66) According to some aspects, the example LECI frame **400A** of FIG. **4A** may be generated by the embedded low-power processor **304** and/or the image processor **306** by assigning higher weights to newer image frames and lower weights to older image frames when computing the weighted average of a sequence of image frames. Accordingly, the example LECI frame of FIG. **4A** illustrates a dog running across a static background, from left to right, with an annotation in the form of a streak effect. The LECI frame of FIG. **4A** shows object **404** annotated differently (i.e., having a darker shade) than objects **402A-E**. In some aspects, the annotation may indicate the object in the LECI frame that corresponds to objects from the most recent image frame in the sequence of image frames that were averaged to generate the LECI frame. Furthermore, the LECI frame of FIG. **4A** shows objects **402A-E** with varying annotations (i.e., having a lighter shade) to indicate that objects **402A-E** correspond to objects from older image frames in the sequence of image frames. In some aspects, object **402A**, object **402B**, object **402C**, object **402D**, and object **402E** may be annotated differently (as shown in FIG. **4A**) to indicate a time progression of objects corresponding to older image frames. For example, object **402A** may correspond to the oldest image frame, object **402B** may correspond to the next oldest image frame, and so on.

(67) In some aspects, other annotations to LECI image frame **400A** are possible. For example, blur effects, added indicators, such as arrows, and/or heat map indications may be added to LECI image frame **400A**. Annotations may also be based on time information from the image frames within the sequence of image frames. Once embedded low-power processor **304** and/or the image processor **306** detect and identify moving objects, such as based on changes in pixels from older to newer

image frames, from image frames within the sequence of image frames, embedded low-power processor **304** and/or the image processor **306** may blur the older objects, add arrows oriented and pointing in the direction from the older objects to the newer objects, and/or heat maps to indicate a time period where the detected objects did not move.

(68) FIG. **4B** illustrates another exemplary long exposure composite image (LECI) frame, according to some aspects of this disclosure. FIG. **4B** illustrates an example LECI frame **400B** that is generated by the embedded low-power processor **304** and/or the image processor **306** by performing moving object detection on a sequence of image frames and computing pixel-wise weighted average over the set of images by assigning higher weights to pixels corresponding to one or more moving objects (e.g., the running dog in FIG. **4B**) in the sequence of images. Similar to the example LECI frame of FIG. **4A**, the example LECI frame **400B** of FIG. **4B** also provides a single-image visual summary of a dog running across a predominantly static background over a period of time (e.g., mountains **416** and trees **418**).

(69) According to some aspects, moving object detection involves identifying and tracking objects that are in motion across a sequence of image frames. Moving object detection may be performed using techniques such as background subtraction, frame differencing, and the like. According to some aspects, the embedded low-power processor **304** implementing a background subtraction method first creates a model of a static background using one or more image frames that are static (i.e., with no moving objects), and moving objects are detected by subtracting the background from one or more subsequent image frames. According to some aspects, the embedded low-power processor **304** implementing a frame differencing method computes the absolute or relative difference between pixel values of corresponding pixels in two or more consecutive image frames to identify regions where pixel values have changed significantly, and the regions of image frames where pixel values have changed indicate moving objects. According to some aspects, embedded low-power processor **304** may compute motion vectors to indicate the motion of pixels between consecutive image frames in the sequence of image frames. According to some aspects, embedded low-power processor **304** may compute the motion vectors using optical flow methods that estimate the apparent displacement of pixels to create vectors that indicate the direction of moving objects.

(70) According to some aspects, once the moving objects in the sequence of image frames are identified, the embedded low-power processor **304** generates a LECI frame by computing the weighted average of a sequence of image frames. According to some aspects, when computing the weighted average of a sequence of image frames, the weights assigned to the pixels may be selected to accentuate one or more moving objects in the sequence of image frames. For example, when performing pixel-wise weighted averaging across a sequence of image frames, pixels corresponding to the moving objects may be assigned higher weights compared to the pixels corresponding to the static background.

(71) According to some aspects, the example LECI frame **400B** of FIG. **4B** may be generated by the embedded low-power processor **304** and/or the image processor **306** by assigning higher weights to pixels corresponding to moving objects (i.e., the running dog) and lower weights to pixels corresponding to the static background and computing the weighted average of a sequence of image frames. Accordingly, in the LECI frame of FIG. **4B**, the images of the object (e.g., objects **412A-F**) have a different annotation (i.e., having a darker shade) than the static background (i.e., the background hills and trees). Other types of annotations may be used to differentiate between a moving object and static objects including highlighting, text annotation, and colorization.

(72) FIG. **5** is a flow diagram for a method **500** for generating long exposure composite image (LECI) frames from stored video data, according to some embodiments. Method **500** can be performed by processing logic that can comprise hardware (e.g., circuitry, dedicated logic, programmable logic, microcode, etc.), software (e.g., instructions executing on a processing device), or a combination thereof. It is to be appreciated that not all steps may be needed to perform

the disclosure provided herein. Further, some of the steps may be performed simultaneously, or in a different order than shown in FIG. 5, as will be understood by a person of ordinary skill in the art.

(73) Method **500** shall first be described with reference to the embodiment of LECI generation module **136** depicted in FIG. 1, although method **500** is not limited to that embodiment.

(74) In **502**, LECI generation module **136** receives a request (e.g., from user **132**) to generate a LECI frame from a video file (e.g., an earlier-recorded video). As noted above, each LECI frame provides a single-image visual summary of activity that occurred over a duration of time. Accordingly, reviewing a LECI frame corresponding to a video segment of a video recording provides a quick and efficient approach to identify activity that may have occurred over the duration of the video segment.

(75) In **504**, the LECI generation module **136** samples a segment of the video file at a predetermined rate to generate a sequence of image frames. According to some aspects, the image-capturing device **300** may generate the video data (i.e., video file). When the image-capturing device **300** is not power-constrained, image-capturing device **300** may capture a video (e.g., video at a high frame rate) of the monitored environment, over a duration of time, and upload the video data (i.e., video file) to media device **104**, content server **120**, and/or a cloud storage system of multimedia environment **102**. According to some aspects, the video file may be an earlier recorded video footage stored on the content server **120**. According to some aspects, the video file may be an earlier recorded video footage stored on a device of the media system **104** and/or the storage module **308** of the image-capturing device **300**. Alternatively, or in addition, the video file may be an earlier recorded video footage stored on cloud storage system of multimedia environment **102**.

(76) According to some aspects, LECI generation module **136** may be configured to sample the video file at a rate based on an amount of activity in the video. According to some aspects, LECI generation module **136** may be configured to quantify activity in the video using techniques such as motion detection, feature extraction, optical flow analysis, machine learning, and the like. The LECI generation module **136** may be configured to sample the video at a rate proportional to the amount of activity in the video.

(77) According to some aspects, LECI generation module **136** determines that a measure of activity corresponding to the video exceeds a predetermined activity threshold, it samples the video to generate image frames at a first predetermined rate. Similarly, when LECI generation module **136** determines that the measure of activity in is below the predetermined activity threshold, LECI generation module **136** may be configured to sample the video to generate image frames at second predetermined rate that is lower than the first predetermined rate. That is the first determined rate may be higher than the second predetermined rate. According to some aspects, the duration of the video segment corresponding to each LECI frame (i.e., the period of activity summarized by each LECI frame) may be varied based on a measure of activity in the video and/or based on selectable user options. In some aspects, the LECI generation module **136** may be configured to vary the period of activity summarized by each LECI frame based on a measure of activity in video.

(78) In **504**, the LECI generation module generates a LECI frame by combining the sequence of image frames. According to some aspects, LECI generation module **136** may generate the LECI frame by computing the weighted average of the sequence of image frames.

(79) According to some aspects, when computing the weighted average of a sequence of image frames, the LECI generation module **136** may select the weights assigned to the pixels may be selected to accentuate one or more moving objects in the monitored environment. For example, when performing pixel-wise weighted averaging across a sequence of image frames, the LECI generation module **136** assigns newer pixels (i.e., pixels corresponding to later sampled image frames) higher weights compared to older pixels (i.e., pixels corresponding to earlier sampled image frames) to obtain a streak effect in the direction of motion in the LECI frame. Alternatively, or in addition, when combining a sequence of image frames to generate a LECI frame, the LECI generation module **136** may modify the pixel colors according to the pixel age such that later image

frames of the sequence may have a color and/or a shade of color that is different from earlier image frames. According to some aspects, when combining a sequence of image frames to generate a LECI frame, the LECI generation module **136** may modify the pixel colors to create a heat map of activity in the LECI frame. According to some aspects, creating a LECI frame that displays a heat map of activity may involve maintaining pixel-change histograms to indicate regions where the most motion has occurred and making them a different color (e.g., using shades of red for high activity areas).

(80) According to some aspects, the LECI generation module **136** may generate a LECI frame by performing moving object detection on a sequence of image frames and computing pixel-wise weighted average over the set of images by assigning higher weights to pixels corresponding to one or more moving objects (e.g., the running dog in FIG. **4B**). According to some aspects, the LECI generation module **136** may identify moving objects in a sequence of image frames and generates a LECI frame by computing the weighted average of a sequence of image frames. When computing the weighted average of a sequence of image frames, the LECI generation module **136** may assign weights to the pixels to accentuate one or more moving objects in the sequence of image frames. For example, when performing pixel-wise weighted averaging across a sequence of image frames, the the LECI generation module **136** assigns higher weights to pixels corresponding to the moving objects compared to the pixels corresponding to the static background.

(81) In **506**, the LECI generation module **136** sends the LECI frame to a network device. For example, the LECI generation module **136** may send the LECI frames to media device **106**. The media device **106** may display an interface for reviewing the received LECI frames. User **132** may review the LECI frame using the interface of media device **106** to identify activity that may have occurred in the video.

(82) FIG. **6** is a flow diagram for a method **600** for generating long exposure composite image (LECI) frames from a sequence of image frames, according to some embodiments. Method **600** can be performed by processing logic that can comprise hardware (e.g., circuitry, dedicated logic, programmable logic, microcode, etc.), software (e.g., instructions executing on a processing device), or a combination thereof. It is to be appreciated that not all steps may be needed to perform the disclosure provided herein. Further, some of the steps may be performed simultaneously, or in a different order than shown in FIG. **6**, as will be understood by a person of ordinary skill in the art.

(83) Method **600** shall first be described with reference to the embodiment of image-capturing device **300** depicted in FIG. **3**, although method **600** is not limited to that embodiment.

(84) In **602**, image-capturing device **300** captures a sequence of image frames over the duration of time. According to some aspects, the sequence of images is captured using an optical sensor **302** of the image-capturing device **300**. According to some aspects, optical sensor **302** generates uncompressed raw pixel data corresponding to a captured image and/or video. Optical sensor **302** may capture color information by using a pattern of red, green, and blue (RGB) filters over the pixels. Alternately, or in addition, optical sensor **302** may capture luminance (Y) and chrominance (UV) components of the pixels. The uncompressed raw pixel data generated by the optical sensor **302** may be sent to the embedded low-power processor **304** and/or image processor **306** for further processing (e.g., quantization and compression).

(85) According to some aspects, optical sensor **302** captures a sequence of image frames at a predetermined frame rate over a duration of time. According to some aspects, the frame rate at which the optical sensor **302** captures the image frames may be predetermined based on a measure of activity in the monitored environment. When monitoring a dynamic environment with a large amount of activity, optical sensor may be configured to record image frames at a higher frame rate to ensure that the sequence of captured image frames captures a sufficient amount of motion. Alternatively, optical sensor **302** may be configured to record image frames at a lower frame rate when monitoring an environment with a low amount of activity. Furthermore, image-capturing device **300** may be configured with software to perform motion detection across the sequence of

captured image frames.

(86) According to some aspects, image-capturing device **300** may quantify a measure of activity in the monitored environment based on comparing two or more of the captured images. For example, image capturing device **300** may quantify activity in the monitored environment may be quantified using techniques such as optical flow method, motion histogram, average displacement vector, mean squared error technique, and the like.

(87) According to some aspects, image-capturing device **300** captures the sequences of image frames at a capture-rate that is proportional to a measure of activity in the monitored environment. Furthermore, image-capturing device **300** may capture the sequence of image frames over a duration that may be inversely proportional to a measure of activity in the monitored environment. When image-capturing device **300** determines that a measure of activity corresponding to the monitored environment exceeds a predetermined activity threshold, optical sensor **302** may be configured to capture the image frames at a first predetermined rate. Similarly, when image-capturing device **300** determines that the measure of activity is below the predetermined activity threshold, optical sensor **302** may be configured to capture the image frames at second predetermined rate that is lower than the first predetermined rate. That is the first determined rate may be higher than the second predetermined rate.

(88) In **604**, image-capturing device **300** generates the LECI frame by combining the sequence of image frames. According to some aspects an embedded low-power processor **304** coupled to optical sensor **302** generates and LECI frame by combining the sequence of image frames. Alternatively, or in addition, image processor **306** of the image-capturing device **300** may generate a LECI frame by combining the sequence of image frames. According to some aspects, embedded low-power processor **304** and/or the image processor **306** may generated the LECI frame by computing the weighted average of the sequence of image frames.

(89) According to some aspects, when computing the weighted average of a sequence of image frames, the embedded low-power processor **304** may select the weights assigned to the pixels may be selected to accentuate one or more moving objects in the monitored environment. For example, when performing pixel-wise weighted averaging across a sequence of image frames, the embedded low-power processor **304** assigns newer pixels higher weights compared to older pixels to obtain a streak effect in the direction of motion in the LECI frame. Alternatively, or in addition, when combining a sequence of image frames to generate a LECI frame, the embedded low-power processor **304** may modify the pixel colors according to the pixel age such that newer image frames of the sequence may have a color and/or a shade of color that is different from older image frames. According to some aspects, when combining a sequence of image frames to generate a LECI frame, the embedded low-power processor **304** may modify the pixel colors to create a heat map of activity in the LECI frame. According to some aspects, creating a LECI frame that displays a heat map of activity may involve maintaining pixel-change histograms to indicate regions where the most motion has occurred and making them a different color (e.g., using shades of red for high activity areas).

(90) According to some aspects, the embedded low-power processor **304** and/or the image processor **306** may generate a LECI frame by performing moving object detection on a sequence of image frames and computing pixel-wise weighted average over the set of images by assigning higher weights to pixels corresponding to one or more moving objects (e.g., the running dog in FIG. 4B). According to some aspects, the embedded low-power processor **304** may identify moving objects in a sequence of image frames and generates a LECI frame by computing the weighted average of a sequence of image frames. When computing the weighted average of a sequence of image frames, the embedded low-power processor **304** may assign weights to the pixels to accentuate one or more moving objects in the sequence of image frames. For example, when performing pixel-wise weighted averaging across a sequence of image frames, the embedded low-power processor **304** assigns higher weights to pixels corresponding to the moving objects

compared to the pixels corresponding to the static background.

(91) In **606**, the image-capturing device **300** transmits, using a transceiver coupled to the embedded low-power processor, the LECI frame to a network device. According to some aspects, the network device may be media device **106** of the media system **104**, content server **120**, and/or a cloud-based storage device. According to some aspects, the image-capturing device **300** may store the LECI frames generated by the embedded low-power processor **304** in storage module **308** and may be periodically transmitted to media device **104** and/or uploaded to a cloud storage system of multimedia environment **102**.

(92) According to some aspects, when the image-capturing device **300** is power-constrained (e.g., low power state, sleep state, or low battery), image-capturing device **300** may not upload LECI frames to the cloud. Instead, image-capturing device **300** may provide an interface for direct access of the LECIs from the storage module **308** over a local network (e.g., user WiFi network).

Alternatively, when the image-capturing device **300** is not power-constrained, image-capturing device **300** may capture a video of the monitored environment over a duration of time, and upload the video data to media device **104** and/or a cloud storage system of multimedia environment **102**. From the uploaded video data, a cloud-based processing module may generate one or more LECIs corresponding to multiple durations of time.

(93) FIG. 7 is a flow diagram for a method **700** for identifying the movement of large objects between frames of video data, according to some embodiments. Method **700** can be performed by processing logic that can comprise hardware (e.g., circuitry, dedicated logic, programmable logic, microcode, etc.), software (e.g., instructions executing on a processing device), or a combination thereof. It is to be appreciated that not all steps may be needed to perform the disclosure provided herein. Further, some of the steps may be performed simultaneously, or in a different order than shown in FIG. 7, as will be understood by a person of ordinary skill in the art.

(94) Method **700** shall first be described with reference to the embodiment of video encoder **312** and central processor **314** depicted in FIG. 3, although method **700** is not limited to that embodiment.

(95) At **702**, video encoder **312** of image-capturing device **300** receives video data from optical sensor **302**. Video encoder **312** may be embedded on a system on chip (SoC) of image capturing device **300**. Video encoder **312** may be configured to receive raw video data from optical sensor **302** and convert it into a digital format that can be streamed, recorded, or transmitted over a network. According to some aspects, video encoder **312** may be configured to compress video data using compression standards such as H.264, H.265 (HEVC), and VP9.

(96) At **704**, video encoder **312** processes a sequence of video frames (e.g., image frames) of the video data to generate a compressed bitstream that includes clusters of motion vectors corresponding to one or more identified movement patterns in the video data. According to some aspects, compared to central processor **314**, video encoder **312** may consume lower power to perform certain video processing operations (e.g., generating motion vectors). According to some aspects, to improve power efficiency, video encoder **312** may be configured to process a sequence of image frames of the video data to generate motion vectors.

(97) According to some aspects, motion vectors may be generated by comparing consecutive video frames in a sequence of video frames. According to some aspects, video encoder **312** may compute the motion vectors using optical flow methods that estimate the apparent displacement of pixels to create vectors that indicate the direction of moving objects. According to some aspects, motion vectors may be generated using a block-matching algorithm. According to some aspects, the motion vectors in the compressed video data generated by video encoder **312** may be scanned by low-power central processor **312** to identify the movement of large objects between frames of video data.

(98) At **706**, motion vectors generated by the video encoder **312** are processed to detect moving objects in a monitored environment corresponding to the video data. Accordingly, large objects in



motion in the video data may be identified in a power-efficient manner by using the on-chip video encoder **312** to compare consecutive video frames and scanning the resulting motion vectors using central processor **312**. According to some aspects, a low-power central processor **314** may identify the large objects in motion by performing motion vector clustering to group motion vectors that may correspond to movements of large objects within a video sequence. In some aspects, the motion vectors may be visualized (e.g., as arrows indicating the direction of movement) on the video data. Central processor **314** may use clustering algorithms such as k-means clustering, hierarchical clustering, and density-based spatial clustering to cluster motion vectors.

#### Example Computer System

(99) Various embodiments may be implemented, for example, using one or more well-known computer systems, such as computer system **700** shown in FIG. 7. For example, the media device **106** may be implemented using combinations or sub-combinations of computer system **700**. Also or alternatively, one or more computer systems **700** may be used, for example, to implement any of the embodiments discussed herein, as well as combinations and sub-combinations thereof.

Computer system **700** may include one or more processors (also called central processing units, or CPUs), such as a processor **704**. Processor **704** may be connected to a communication infrastructure or bus **706**.

(100) Computer system **700** may also include user input/output device(s) **703**, such as monitors, keyboards, pointing devices, etc., which may communicate with communication infrastructure **706** through user input/output interface(s) **702**.

(101) One or more of processors **704** may be a graphics processing unit (GPU). In an embodiment, a GPU may be a processor that is a specialized electronic circuit designed to process mathematically intensive applications. The GPU may have a parallel structure that is efficient for parallel processing of large blocks of data, such as mathematically intensive data common to computer graphics applications, images, videos, etc.

(102) Computer system **700** may also include a main or primary memory **708**, such as random access memory (RAM). Main memory **708** may include one or more levels of cache. Main memory **708** may have stored therein control logic (i.e., computer software) and/or data.

(103) Computer system **700** may also include one or more secondary storage devices or memory **710**. Secondary memory **710** may include, for example, a hard disk drive **712** and/or a removable storage device or drive **714**. Removable storage drive **714** may be a floppy disk drive, a magnetic tape drive, a compact disk drive, an optical storage device, tape backup device, and/or any other storage device/drive.

(104) Removable storage drive **714** may interact with a removable storage unit **718**. Removable storage unit **718** may include a computer usable or readable storage device having stored thereon computer software (control logic) and/or data. Removable storage unit **718** may be a floppy disk, magnetic tape, compact disk, DVD, optical storage disk, and/or any other computer data storage device. Removable storage drive **714** may read from and/or write to removable storage unit **718**.

(105) Secondary memory **710** may include other means, devices, components, instrumentalities or other approaches for allowing computer programs and/or other instructions and/or data to be accessed by computer system **700**. Such means, devices, components, instrumentalities or other approaches may include, for example, a removable storage unit **722** and an interface **720**. Examples of the removable storage unit **722** and the interface **720** may include a program cartridge and cartridge interface (such as that found in video game devices), a removable memory chip (such as an EPROM or PROM) and associated socket, a memory stick and USB or other port, a memory card and associated memory card slot, and/or any other removable storage unit and associated interface.

(106) Computer system **700** may further include a communication or network interface **724**. Communication interface **724** may enable computer system **700** to communicate and interact with any combination of external devices, external networks, external entities, etc. (individually and

collectively referenced by reference number **728**). For example, communication interface **724** may allow computer system **700** to communicate with external or remote devices **728** over communications path **726**, which may be wired and/or wireless (or a combination thereof), and which may include any combination of LANs, WANs, the Internet, etc. Control logic and/or data may be transmitted to and from computer system **700** via communication path **726**.

(107) Computer system **700** may also be any of a personal digital assistant (PDA), desktop workstation, laptop or notebook computer, netbook, tablet, smart phone, smart watch or other wearable, appliance, part of the Internet-of-Things, and/or embedded system, to name a few non-limiting examples, or any combination thereof.

(108) Computer system **700** may be a client or server, accessing or hosting any applications and/or data through any delivery paradigm, including but not limited to remote or distributed cloud computing solutions; local or on-premises software (“on-premise” cloud-based solutions); “as a service” models (e.g., content as a service (CaaS), digital content as a service (DCaaS), software as a service (SaaS), managed software as a service (MSaaS), platform as a service (PaaS), desktop as a service (DaaS), framework as a service (FaaS), backend as a service (BaaS), mobile backend as a service (MBaaS), infrastructure as a service (IaaS), etc.); and/or a hybrid model including any combination of the foregoing examples or other services or delivery paradigms.

(109) Any applicable data structures, file formats, and schemas in computer system **700** may be derived from standards including but not limited to JavaScript Object Notation (JSON), Extensible Markup Language (XML), Yet Another Markup Language (YAML), Extensible Hypertext Markup Language (XHTML), Wireless Markup Language (WML), MessagePack, XML User Interface Language (XUL), or any other functionally similar representations alone or in combination. Alternatively, proprietary data structures, formats or schemas may be used, either exclusively or in combination with known or open standards.

(110) In some embodiments, a tangible, non-transitory apparatus or article of manufacture comprising a tangible, non-transitory computer useable or readable medium having control logic (software) stored thereon may also be referred to herein as a computer program product or program storage device. This includes, but is not limited to, computer system **700**, main memory **708**, secondary memory **710**, and removable storage units **718** and **722**, as well as tangible articles of manufacture embodying any combination of the foregoing. Such control logic, when executed by one or more data processing devices (such as computer system **700** or processor(s) **704**), may cause such data processing devices to operate as described herein.

(111) Based on the teachings contained in this disclosure, it will be apparent to persons skilled in the relevant art(s) how to make and use embodiments of this disclosure using data processing devices, computer systems and/or computer architectures other than that shown in FIG. 7. In particular, embodiments can operate with software, hardware, and/or operating system implementations other than those described herein.

## CONCLUSION

(112) It is to be appreciated that the Detailed Description section, and not any other section, is intended to be used to interpret the claims. Other sections can set forth one or more but not all exemplary embodiments as contemplated by the inventor(s), and thus, are not intended to limit this disclosure or the appended claims in any way.

(113) While this disclosure describes exemplary embodiments for exemplary fields and applications, it should be understood that the disclosure is not limited thereto. Other embodiments and modifications thereto are possible, and are within the scope and spirit of this disclosure. For example, and without limiting the generality of this paragraph, embodiments are not limited to the software, hardware, firmware, and/or entities illustrated in the figures and/or described herein. Further, embodiments (whether or not explicitly described herein) have significant utility to fields and applications beyond the examples described herein.

(114) Embodiments have been described herein with the aid of functional building blocks

illustrating the implementation of specified functions and relationships thereof. The boundaries of these functional building blocks have been arbitrarily defined herein for the convenience of the description. Alternate boundaries can be defined as long as the specified functions and relationships (or equivalents thereof) are appropriately performed. Also, alternative embodiments can perform functional blocks, steps, operations, methods, etc. using orderings different than those described herein.

(115) References herein to “one embodiment,” “an embodiment,” “an example embodiment,” or similar phrases, indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it would be within the knowledge of persons skilled in the relevant art(s) to incorporate such feature, structure, or characteristic into other embodiments whether or not explicitly mentioned or described herein. Additionally, some embodiments can be described using the expression “coupled” and “connected” along with their derivatives. These terms are not necessarily intended as synonyms for each other. For example, some embodiments can be described using the terms “connected” and/or “coupled” to indicate that two or more elements are in direct physical or electrical contact with each other. The term “coupled,” however, can also mean that two or more elements are not in direct contact with each other, but yet still co-operate or interact with each other.

(116) The breadth and scope of this disclosure should not be limited by any of the above-described exemplary embodiments, but should be defined only in accordance with the following claims and their equivalents.

## Claims

1. A computer-implemented method for generating a long exposure composite image (LECI) frame, wherein the LECI frame provides a summary of activity in a monitored environment over a duration of time, comprising: capturing, by at least one computer processor of an image-capturing device, a sequence of image frames over the duration of time using an optical sensor of the image-capturing device; generating, using an embedded low-power processor coupled to the optical sensor and dedicated to generating LECI frames, the LECI frame by combining the sequence of image frames; and transmitting, using a transceiver coupled to the embedded low-power processor, the LECI frame to a network device.
2. The computer-implemented method of claim 1, wherein the sequence of image frames comprises a first set of image frames and a second set of image frames, wherein the first set of image frames is associated with a first time period within the duration of time and the second set of image frames is associated with a second time period within the duration of time, wherein the first time period is more recent than the second time period, and wherein generating the LECI frame comprises: performing a weighted averaging of the sequence of image frames by: assigning a first set of weights to the first set of image frames of the sequence of image frames; and assigning a second set of weights to the second set of image frames, wherein each of the first set of weights is greater than all of the second set of weights.
3. The computer-implemented method of claim 1, wherein generating the LECI frame comprises: combining the sequence of image frames by computing a pixel-wise weighted average of pixel values across the sequence of image frames.
4. The computer-implemented method of claim 1, wherein generating the LECI frame comprises: identifying a moving object in one or more image frames of the sequence of image frames; and performing the weighted averaging by assigning higher weights to pixels corresponding to the moving object in the one or more image frames.

5. The computer-implemented method of claim 1, wherein generating the LECI frame comprises: identifying relative displacement of one or more pixels between a set of image frames of the sequence of image frames to generate one or more direction vectors indicating direction of motion of an object in the sequence of image frames.
6. The computer-implemented method of claim 1, wherein capturing the sequence of image frames comprises: capturing the sequence of image frames at a capture-rate that is proportional to a measure of activity in the monitored environment.
7. The computer-implemented method of claim 1, wherein the duration of time is inversely proportional to a measure of activity in the monitored environment.
8. The computer-implemented method of claim 1, wherein the image-capturing device is a battery-operated camera.
9. A system for generating a long exposure composite image (LECI) frame, wherein the LECI frame provides a summary of activity in a monitored environment over a duration of time, comprising: one or more memories; and at least one processor coupled to at least one of one or more memories of an image-capturing device and configured to perform operations comprising: capturing a sequence of image frames over the duration of time using an optical sensor of the image-capturing device; generating, using an embedded low-power processor coupled to the optical sensor and dedicated to generating LECI frames, the LECI frame by combining the sequence of image frames; and transmitting, using a transceiver coupled to the embedded low-power processor, the LECI frame to a network device.
10. The system of claim 9, wherein the sequence of image frames comprises a first set of image frames and a second set of image frames, wherein the first set of image frames is associated with a first time period within the duration of time and the second set of image frames is associated with a second time period within the duration of time, wherein the first time period is more recent than the second time period, and wherein generating the LECI frame comprises: performing a weighted averaging of the sequence of image frames by: assigning a first set of weights to the first set of image frames of the sequence of image frames; and assigning a second set of weights to the second set of image frames, wherein each of the first set of weights is greater than all of the second set of weights.
11. The system of claim 9, wherein generating the LECI frame comprises: combining the sequence of image frames by computing a pixel-wise weighted average of pixel values across the sequence of image frames.
12. The system of claim 9, wherein generating the LECI frame comprises: identifying a moving object in one or more image frames of the sequence of image frames; and performing a weighted averaging by assigning higher weights to pixels corresponding to the moving object in the one or more image frames.
13. The system of claim 9, wherein generating the LECI frame comprises: identifying relative displacement of one or more pixels between a set of image frames of the sequence of image frames to generate one or more direction vectors indicating direction of motion of an object in the sequence of image frames.
14. The system of claim 9, wherein capturing the sequence of image frames comprises: capturing the sequence of image frames at a capture-rate that is proportional to a measure of activity in the monitored environment.
15. The system of claim 9, wherein the duration of time is inversely proportional to a measure of activity in the monitored environment.
16. A non-transitory computer-readable medium having instructions stored thereon that, when executed by at least one computing device, cause the at least one computing device to perform operations for generating a long exposure composite image (LECI) frame, wherein the LECI frame provides a summary of activity in a monitored environment over a duration of time, the operations comprising: capturing, by at least one computer processor of an image-capturing device, a

sequence of image frames over the duration of time using an optical sensor of the image-capturing device; generating, using an embedded low-power processor coupled to the optical sensor and dedicated to generating LECI frames, the LECI frame by combining the sequence of image frames; and transmitting, using a transceiver coupled to the embedded low-power processor, the LECI frame to a network device.

17. The non-transitory computer-readable medium of claim 16, wherein the sequence of image frames comprises a first set of image frames and a second set of image frames, wherein the first set of image frames is associated with a first time period within the duration of time and the second set of image frames is associated with a second time period within the duration of time, wherein the first time period is more recent than the second time period, and wherein generating the LECI frame comprises: performing a weighted averaging of the sequence of image frames by: assigning a first set of weights to the first set of image frames of the sequence of image frames; and assigning a second set of weights to the second set of image frames, wherein each of the first set of weights is greater than all of the second set of weights.

18. The non-transitory computer-readable medium of claim 16, wherein generating the LECI frame comprises: combining the sequence of image frames by computing a pixel-wise weighted average of pixel values across the sequence of image frames.

19. The non-transitory computer-readable medium of claim 16, wherein generating the LECI frame comprises: identifying a moving object in one or more image frames of the sequence of image frames; and performing the weighted averaging by assigning higher weights to pixels corresponding to the moving object in the one or more image frames.

20. The non-transitory computer-readable medium of claim 16, wherein capturing the sequence of image frames comprises: capturing the sequence of image frames at a capture-rate that is proportional to a measure of activity in the monitored environment.

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